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Multiway cuts in node weighted graphs[☆]

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Abstract

A $(2 - 2/k)$ approximation algorithm is presented for the node multiway cut problem, thus matching the result of Dahlhaus et al. (SIAM J. Comput. 23 (4) (1994) 864–894) for the edge version of this problem. This is done by showing that the associated LP-relaxation always has a half-integral optimal solution.

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1. Introduction

Finding a minimum s – t cut in a graph occupies a central place in combinatorial optimization. The edge multiway cut problem is its generalization in which a given set of k terminals need to be disconnected by deleting a minimum weight set of edges. This problem was shown to be NP-hard and MAX SNP-hard even for fixed $k \geq 3$ by Dahlhaus et al. [4]. They also gave a $(2 - 2/k)$ approximation algorithm for this problem, using the notion of *isolating cuts*. Calinescu et al. [2] improved this to a $1.5 - 1/k$ approximation through a novel geometric relaxation of the problem. This was further improved by Karger et al. [9] to a $1.3438 - \epsilon_k$ approximation.

In this paper, we consider *the node multiway cut problem*, in which a given set of terminals needs to be disconnected by removing a minimum weight set of nodes from the graph. It is easy to see that the edge version reduces to the node version via an

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approximation preserving reduction. However, no reductions are known in the reverse direction, and we believe that the node version is strictly harder.

The technique of finding isolating cuts [4] or the embedding ideas [2] do not extend to give a constant factor approximation algorithm for the node version. Our main result is that we can still obtain a $(2 - 2/k)$ approximation algorithm for this version. The algorithm follows from proving that the LP-relaxation of the integer programming formulation of this problem always has a half-integral optimal solution.

The node multiway cut problem reduces to the directed multiway cut problem via an approximation preserving reduction; again a reduction in the other direction is not known. It is natural then to ask if the directed version also exhibits this half-integrality property. We give an example to show that it does not.

We also give an approximation preserving reduction from the node cover problem to the node multiway cut problem, in case k is not fixed. Since the best known approximation factor for the node cover problem is 2, improving our approximation guarantee will not be straightforward.

We start by giving definitions and describing simple reductions mentioned above in Section 2. In this section, we also show that the directed version is NP-hard and MAX SNP-hard [11] even for two terminals. The main results are presented in Section 3.

This paper first appeared in [8]. In that version, we had also given a preliminary factor $O(\log k)$ approximation algorithm for the directed version. Recently, Naor and Zosin have obtained a 2 approximation algorithm for this case, using a new LP formulation [10].

2. Definitions and reductions

Let $G = (V, E)$ be an undirected or directed graph with node weights, $w : (V - T) \rightarrow \mathbf{R}^+$, where T is the set of k terminals, $s_1, s_2, \dots, s_k$. A *node multiway cut* is a set of non-terminal nodes whose deletion disconnects each terminal from all the others, i.e., leaves no path from any terminal to any other terminal. The graph has a node multiway cut only if the terminals form an independent set. The node multiway cut problem is to find such a cut of minimum weight. In the special case that all the weights are equal to 1, we want to minimize the cardinality of the cut; we refer to this as the *unweighted* problem. In the *edge multiway cut* problem, it is the edges that have weights instead of the nodes and the problem is again to find a minimum weight set of edges whose deletion disconnects each terminal from all the others.

The version of the problem in which the terminal nodes can also be included in the multiway cut can be mapped to our version by introducing, for each terminal s_i , a new node t_i and edge (s_i, t_i) . The set $t_1, t_2, \dots, t_k$ is the new set of terminals and it is easy to see that there is a one-to-one correspondence between node multiway cuts which could include terminals in the original instance and node multiway cuts which do not include a terminal in the new instance.

The multiway cut problem has four different versions depending on whether the underlying graph is undirected or directed and whether the cut is an edge or a node cut. Figure 1 illustrates the relationships between these versions. An arrow in Fig. 1 from version A to version B implies that an instance of version A can be reduced in

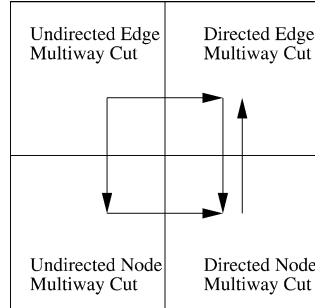


Fig. 1. The four versions of multiway cuts and reductions between them.

polynomial time to an instance of version B via an approximation preserving reduction. All these reductions apply to both the unweighted and weighted settings and preserve the number of terminals. We omit proofs for these reductions since they are straightforward and employ standard techniques. These reductions, together with the fact that the unweighted, undirected, edge-multiway cut problem is NP-hard even for $k = 3$ [4] implies the NP-hardness of the various versions for $k \geq 3$.

In the special case of trees, it is easy to show that the node multiway cut problem can be solved in polynomial time by a straightforward dynamic programming algorithm.

In contrast to the undirected versions, which are NP-hard for $k \geq 3$, though polynomial time solvable for $k = 2$ (by network flow techniques), the directed versions are NP-hard even for $k = 2$.

Theorem 1. *The problem of computing the minimum edge (node) multiway cut in directed graphs is NP-hard and MAX SNP-hard for all $k \geq 2$, even if all edge (node) weights are 1.*

Proof. From the discussion preceding this theorem we know that the unweighted undirected node 3-way cut problem is NP-complete. We show how this problem can be reduced to the weighted directed node 2-way cut problem. The weights will be polynomially bounded and hence can be removed in the standard way by replacing weighted nodes by cliques. The result for the edge version follows since in directed graphs node multiway cuts can be reduced to edge multiway cuts.

Let $G = (V, E)$ be an undirected graph with 3 distinguished terminals a, b, c , and with all node weights 1. Construct a directed graph D as follows: replace every edge of G by two

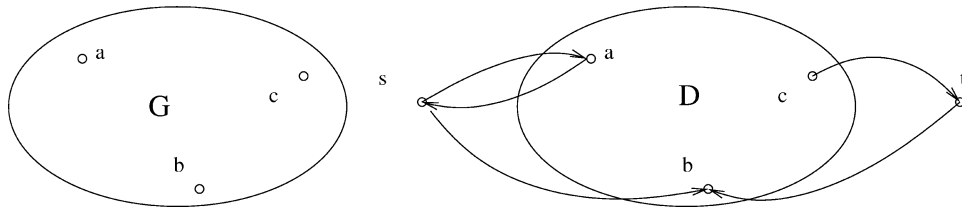


Fig. 2. The construction.

opposite arcs, add two new nodes s, t , which are the new terminals, and add the following arcs: $s \rightarrow a, s \rightarrow b, a \rightarrow s, c \rightarrow t, t \rightarrow b$. Nodes a, b, c are given a large weight M , e.g., $M = 2n$, and the rest of the nodes of G have weight 1. Clearly, an optimal 2-way cut Q in D will not pick any of the heavy nodes. It is easy to see from the construction that, for any set Q of nodes, not containing a, b, c, s , and t , deleting Q from G leaves a path between two of the nodes a, b, c if and only if deleting Q from D leaves a path from s to t or a path from t to s . Thus, an optimal 2-way cut in D corresponds to an optimal 3-way cut in G . $\square$

The multiway cut problems are a special case of the *multicut* problem. In that problem we are given a graph $G = (V, E)$ with positive weights on its nodes or edges, and a set of pairs of ‘source’–‘sink’ nodes $(s_1, t_1) \dots (s_k, t_k)$, and we wish to delete a minimum weight set of nodes or edges so that the resulting graph does not contain a path from any source s_i to its corresponding sink t_i . The (node or edge) multiway cut problem for a set T of terminals coincides with the (node or edge) multicut problem where the set of source–sink pairs consists of all pairs of distinct nodes of T . The edge multicut problem in undirected graphs was studied in [7] which gave a $O(\log k)$ approximation algorithm and also proved a $O(\log k)$ relationship to the corresponding maximum multicommodity flow problem; these results extend also to the node multicut problem for undirected graphs. In the directed case we do not know of any nontrivial approximation algorithms in general (i.e., better than the trivial $O(k)$ factor). However, the multiway cut problem is easier than the more general multicut problem as one can obtain a constant factor approximation for the edge multiway cut for all k [2–4,9].

3. Undirected node weighted graphs

Let $G = (V, E)$ be an undirected graph with node weights, $w : (V - T) \rightarrow \mathbf{R}^+$, where T is the set of k terminals, $s_1, s_2, \dots, s_k$. We assume, without loss of generality, that the terminals form an independent set.

In the case of 2 terminals the problem of course coincides with the classical min-cut problem, and can be solved in polynomial time. For $k = 3$ or more terminals, as in the case of the edge multiway cut problem, the node multiway cut problem is NP-hard and MAX SNP-hard. We now show that the problem for general k is likely to be even harder to approximate, since it can be no easier to approximate than the node cover problem.

Theorem 2. *An α -approximation algorithm for the weighted (respectively unweighted) minimum node multiway cut problem can be used to obtain an α -approximation for the weighted (respectively unweighted) minimum node cover problem.*

Proof. Given a graph $G = (V, E)$ with node-weights, $w : V \rightarrow \mathbf{R}^+$, we construct another graph G' by adding a new node v' for each node $v \in V$ and the edge (v', v) . The new nodes added are the set of terminals that we wish to disconnect. It is now easy to see that a node multiway cut in G' is a node cover in G and vice-versa. $\square$

The node cover problem can be approximated with ratio 2 in polynomial time; it is a long-standing open problem to find an algorithm with better approximation ratio (i.e., a constant $r < 2$, independent of the graph) or prove that it is not possible. Hence, achieving a ratio better than 2 for the node multiway cut problem will be at least as difficult.

The $(2 - 2/k)$ approximation algorithm for the edge multiway cut problem [4] does not extend to node multiway cuts as we explain now. Define an *isolating cut* for terminal s_i to be a cut that separates s_i from the rest of the terminals. A minimum isolating cut for s_i can be computed in polynomial time by identifying the remaining terminals, and finding a minimum cut separating them from s_i . The algorithm of [4] finds such cuts for each terminal, discards the heaviest cut, and picks the union of the remaining cuts. The approximation factor is proven by observing that on doubling each edge in the optimal multiway cut, we can partition these edges into k isolating cuts, one for each terminal. Each of these cuts must be at least as heavy as the corresponding isolating cut found by the algorithm, hence giving the approximation factor. Isolating cuts can be defined appropriately and found efficiently for our setting of node weighted graphs as well. However, the second part of the argument crucially rests on the fact that each edge has two end points, and hence contributes to two isolating cuts. This does not carry over to the node case. In fact, the following example shows that this algorithm achieves a factor of no better than $(k - 1)$: let G be a graph with $2k + 1$ nodes, $s_1, \dots, s_k, u_1, \dots, u_k, v$. v has weight 1, and is connected to each u_i . Each u_i has weight $1 - \epsilon$ for a small constant $\epsilon > 0$, and is connected to s_i . Then, the optimal node multiway cut is just v , and has weight 1. However, the best isolating cut for each terminal s_i consists of the node u_i ; thus, the algorithm finding isolating cuts will pick $k - 1$ of the u_i 's, and will incur a cost of $(k - 1)(1 - \epsilon)$.

Calinescu et al. [2] use a stronger LP formulation to obtain a better approximation for the edge multiway cut. This LP relaxation is obtained by noting that if all edges of a multiway cut are assigned length 1 (and other edges a length 0) then the sum of the distances of any node to the k terminals is exactly $k - 1$. In the case of node multiway cuts the lengths have to be associated with the nodes and then the above observation is no longer valid.

Our approach is to consider the LP relaxation of the node multiway cut problem and to work with its dual which is a multicommodity flow problem.

3.1. A multicommodity flow problem

With every pair of terminals s_i, s_j we associate a commodity and designate one of s_i, s_j as the source and the other as the sink for this commodity. Thus the total number of commodities is $k(k - 1)/2$. The multicommodity flow problem is to maximize the sum of the flows of the commodities subject to flow conservation (a commodity is conserved at each node except the source and sink for that commodity) and throughput constraints (the sum of the flows of all commodities through a node cannot exceed the weight of the node).

The throughput and conservation constraints can be formulated as linear inequalities and the objective function, which is the sum of flows of all commodities is also linear. Thus, it is straight-forward to formulate this multicommodity flow problem as a linear program with $O(k^2 m)$ variables and constraints, where m is the number of edges. An optimal solution to the LP can be found in polynomial time.

Another way of writing a linear program for this problem, which although it has exponentially many variables, is interesting from the viewpoint of multiway cuts is as follows: For each path p_i between two distinct terminals, let f_i be a variable denoting the flow along this path. We need to ensure that the sum of the flows along paths that go through a node does not exceed the weight of the node, i.e.,

$$\sum_{i: v \in p_i} f_i \leq w_v, \quad v \in V - T.$$

Thus the linear program is

$$\begin{aligned} \text{P1:} \quad & \text{maximize} \quad \sum_i f_i \\ & \text{subject to} \quad \sum_{i: v \in p_i} f_i \leq w_v, \quad \forall v \in V - T, \\ & \quad \quad \quad f_i \geq 0, \quad \forall \text{ path } p_i, \end{aligned}$$

and its dual is

$$\begin{aligned} \text{P2:} \quad & \text{minimize} \quad \sum_{v \in V - T} d_v w_v \\ & \text{subject to} \quad \sum_{v \in p_i} d_v \geq 1, \quad \forall \text{ path } p_i, \\ & \quad \quad \quad d_v \geq 0, \quad \forall v \in V - T. \end{aligned}$$

The dual can be viewed as an assignment of non-negative ‘length’ (or ‘distance’) labels, d_v , to the non-terminal nodes. These labels induce a distance function between pairs of nodes. Define the *length* of a path to be the sum of the labels of the nodes along the path. The *distance* between two nodes is the length of the shortest path between them. The constraints of the dual LP imply that the distance between any two terminals is at least 1; the objective is to minimize $\sum_{v \in V - T} d_v w_v$. Note that we never need to assign any node a length label more than 1. Hence, any integral solution to this LP is a 0–1 assignment of length labels to the nodes such that each path, p_i , contains at least one node with $d_v = 1$. Thus, any integral solution to the dual LP corresponds to a node multiway cut in G which is of weight equal to the value of the solution. The converse is also true, i.e., every node multiway cut in G yields an integral solution to the dual LP of value equal to the weight of the multiway cut. This correspondence between the node multiway cuts in G and integral solutions to the dual linear program implies that the optimal solution to this linear program, which by the Duality theorem of linear programming is equal to the maximum multicommodity flow, is a lower bound on the value of the minimum node multiway cut.²

Although the dual LP has exponential size, it can be solved in polynomial time. The standard way is to take the polynomial size LP formulation of the maximum multicommodity flow problem that we mentioned earlier, construct its dual (which is

² This also follows directly from the fact that the multiway cut acts as a bottleneck to flow and hence the value of the flow cannot exceed the capacity of any multiway cut.

polynomially large) and solve it. This polynomial size dual LP includes, besides the variables d_v , a variable $y_{u,j}$, for every node u and each $j = 1, \dots, k$, which is supposed to represent the distance between terminal s_j and node u . The reformulated dual LP is as follows

$$\begin{aligned} \text{P3: } & \text{minimize} && \sum_{v \in V-T} d_v w_v \\ & \text{subject to} && \\ & && y_{v,j} \leq y_{u,j} + d_v, \quad \forall (u, v) \in E, \quad \forall j = 1, \dots, k, \\ & && y_{s_j,j} = 0, \quad \forall j = 1, \dots, k, \\ & && y_{s_i,j} \geq 1, \quad \forall i, j = 1, \dots, k; \quad i \neq j, \\ & && d_v \geq 0, \quad \forall v \in V - T. \end{aligned}$$

In the first constraint above, if v is a terminal, we let $d_v = 0$ in the constraint. It is easy to see that the LP's P2 and P3 are equivalent. Given a feasible solution $\{d_v: v \in V - T\}$ to P2, we can extend it to a solution of P3 by letting $y_{u,j}$ be the distance from s_j to u according to the node lengths d_v . Conversely, given any feasible solution to P3, the variables $y_{u,j}$ must be upper bounded by the corresponding distances because of the first and second constraints; the third constraints ensure then that any two terminals are at least distance 1 apart and hence the variables d_v form a solution to P2.

Is maximum multicommodity flow equal to the minimum multiway cut? The example in Fig. 3 shows that this is not the case. The graph in this example has, besides the k terminals, k nodes of unit weight and a center node of weight k . The maximum multicommodity flow routes a total flow of $k/2$ while the weight of the minimum node multiway cut is $k - 1$.

The optimal solution to the dual LP for this example assigns a length label 0 to the center node and labels of $1/2$ to the other non-terminal nodes; each pair of terminals is a unit distance apart and the objective function has value $k/2$ which is equal to the maximum multicommodity flow. For multiway cuts, the example of Fig. 3 is really the worst that things get, in the sense that the dual linear program always has an optimal solution that is half-integral.

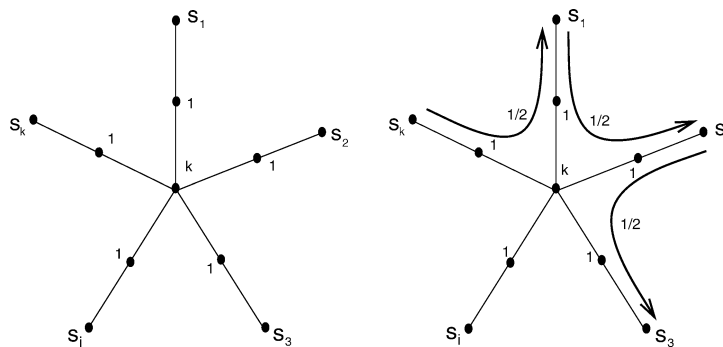


Fig. 3. An example to show that maximum flow is not equal to the minimum node multiway cut.

3.2. Half-integrality of the optimum

Consider the optimal solutions to the primal and dual linear programs. Let $d: (V - T) \rightarrow \mathbf{R}^+$ be the assignment of length labels corresponding to this solution. By the Duality theorem, $\sum_{v \in V-T} d_v w_v$ is equal to the maximum multicommodity flow.

Note that the terminals have no length labels but for the purpose of computing distances we assign them a label 0. A *flow path* is a path between two distinct terminals such that there is a non-zero flow along this path. A node is *saturated* if the total flow through it equals its weight. The duality theory of linear programming (see [1]) says that the constraints $(\sum_{i: v \in p_i} f_i \leq w_v)$, $(d_v \geq 0)$ and $(\sum_{v \in p_i} d_v \geq 1)$, $(f_i \geq 0)$ form complementary pairs such that at optimality at most one constraint in each pair can be slack, i.e., not satisfied as an equality. This implies that

- (1) a node that has a non-zero length label must be saturated;
- (2) any flow path must have length exactly one.

Any two terminals are at least a unit distance apart under the length assignment d . We use this fact and the two complementary slackness conditions to show that there is an optimal solution to the dual LP that is half integral.

With each terminal s_i we associate a *region*, S_i , which is the set of nodes at a zero distance from s_i . The *boundary* of this region is the set of nodes adjacent to this region, and is denoted by $\Gamma(S_i)$. By definition, all nodes v of the boundary $\Gamma(S_i)$ have $d_v > 0$. Since any two terminals are at least a unit distance apart, the regions $S_1, S_2, \dots, S_k$ are disjoint. Further, the boundary of a region does not have a node that belongs to any region.

The boundaries of the regions are not necessarily disjoint. If v is a node common to the boundary of two regions, S_i, S_j , then there is a path between terminals s_i and s_j on which v is the only node with a non-zero length label; hence $d_v = 1$. Let $M = \bigcup_{i=1}^k \Gamma(S_i)$ be the set of nodes in the boundaries and let M^1 be the nodes in M that belong to the boundaries of at least two regions. Then by the previous argument, $\forall v \in M^1: d_v = 1$; the superscript in M^1 denotes the fact that all nodes in this set have a length label 1. We next show that there is an optimal solution to the dual LP in which the remaining nodes of M , i.e., $M - M^1$, have a length label $1/2$ (Theorem 4). We denote this set of nodes by $M^{1/2} = M - M^1$; every node in this set belongs to the boundary of exactly one region.

Lemma 3. Any flow path from s_i to s_j either uses exactly one node from M^1 or it uses exactly two nodes from $M^{1/2}$.

Proof. Let v_i, v_j be the first and last nodes from M on this flow path. Clearly, $v_i \in \Gamma(S_i)$ and $v_j \in \Gamma(S_j)$. We have two cases

- (1) $v_i = v_j$. Then $v_i = v_j \in M^1$. Further, the definition of v_i, v_j implies that this is the only node of M on this path.
- (2) $v_i \neq v_j$. We first show that $v_i, v_j \in M^{1/2}$. Suppose $v_i \in M^1$. Then $d_{v_i} = 1$ and since $d_{v_j} > 0$ the length of this flow path is strictly greater than 1, contradicting

the second complementary slackness condition. For the same reason $v_j \notin M^1$ and so $v_i, v_j \in M - M^1 = M^{1/2}$.

We now show that v_i, v_j are the only nodes from M on this flow path. Let $v_k \in \Gamma(S_k)$ be another node from M on this path. Since $v_k \in \Gamma(S_k)$, there is a path from terminal s_k to node v_k , all nodes on which—with the exception of v_k —have a length label 0. This coupled with the facts that the length of the flow path $s_i - v_i - v_k - v_j - s_j$ is exactly one and $d_{v_i}, d_{v_j} > 0$ implies that the paths $s_i - v_i - v_k - s_k$ and $s_k - v_k - v_j - s_j$ have length strictly less than 1. Since k is distinct from at least one of i and j , we get a pair of distinct terminals (s_i, s_k or s_j, s_k) the distance between which is less than one; a contradiction. $\square$

Let W_S denote the sum of the weights of the nodes in set S , i.e., $W_S = \sum_{v \in S} w_v$.

Theorem 4. *For any assignment of non-negative weights to the nodes of G there exists an optimal solution to the dual LP that is half integral. Moreover, given any optimal solution to the dual LP we can obtain, from it, a half-integral optimal solution in linear time.*

Proof. We start with an arbitrary optimal solution to the dual LP, and construct a feasible solution that is half-integral and has the same value. Note that by the Duality theorem, this is equivalent to showing that the half-integral solution has value equal to the maximum flow.

Corresponding to the optimal solution to the dual LP, compute the sets $S_i, \Gamma(S_i)$ for $1 \leq i \leq k$, and the sets $M, M^1, M^{1/2}$ as defined above. Since all nodes of M have a non-zero length label, they are saturated (the first complementary slackness condition). This together with Lemma 3 implies that the total flow carried by all flow paths using one node of M^1 equals W_{M^1} and total flow carried by all flow paths using two nodes of $M^{1/2}$ is $\frac{1}{2}W_{M^{1/2}}$. Hence,

$$\text{maximum flow} = W_{M^1} + \frac{1}{2}W_{M^{1/2}}. \quad (1)$$

Now consider the following assignment of length labels to the nodes

$$d_v^* = \begin{cases} 1 & \text{if } v \in M^1, \\ \frac{1}{2} & \text{if } v \in M^{1/2}, \\ 0 & \text{otherwise.} \end{cases}$$

Any path from s_i to s_j must use nodes from both $\Gamma(S_i)$ and $\Gamma(S_j)$. Let $v_i \in \Gamma(S_i)$ and $v_j \in \Gamma(S_j)$ be two nodes on this path. If $v_i = v_j$ then $v_i \in M^1$ and $d_{v_i}^* = 1$. Else if $v_i \neq v_j$ then $d_{v_i}^*, d_{v_j}^* \geq 1/2$. In either case the length of this path is at least one. Hence d^* is a feasible solution to the dual LP.

For this length label assignment the value of the objective function is

$$\sum_{v \in V} d_v^* w_v = W_{M^1} + \frac{1}{2}W_{M^{1/2}}.$$

But this is equal to the maximum flow (Eq. 1). Hence, d^* is an optimal solution to the dual LP. $\square$

3.3. Approximate max-flow min-multiway cut theorem

Theorem 5 (Approximate max-flow min-multiway cut theorem).

$$\text{maximum flow} \leq \text{minimum node multiway cut} \leq \left(2 - \frac{2}{k}\right) \cdot \text{maximum flow}.$$

Proof. Let d be a half-integral optimal solution to the dual LP. Hence,

$$\text{maximum flow} = \sum_{v \in V-T} d_v w_v = W_{M^1} + \frac{1}{2} W_{M^{1/2}}.$$

Since any path between two terminals contains at least one node from M , the set of nodes, M , forms a multiway cut. Further, since each node of M has a length label that is $1/2$ or 1 , the sum of the weights of these nodes is at most $2 \sum_{v \in M} d_v w_v = 2 \sum_{v \in V-T} d_v w_v$, which is twice the maximum flow.

We can prove a better bound on the weight of the multiway cut by discarding a part of the boundary of one of the regions. Recall that every node in $M^{1/2}$ belongs to the boundary of only one region. Let $\Gamma^{1/2}(S_i) = M^{1/2} \cap \Gamma(S_i)$ be the set of nodes that are unique to the boundary of the region S_i . The sets $\Gamma^{1/2}(S_1), \Gamma^{1/2}(S_2), \dots, \Gamma^{1/2}(S_k)$ form a partition of the set $M^{1/2}$ and if $\Gamma^{1/2}(S_j)$ is the heaviest set in the partition then

$$W_{\Gamma^{1/2}(S_j)} \geq \frac{1}{k} W_{M^{1/2}}.$$

Claim 6. $M - \Gamma^{1/2}(S_j)$ is a node multiway cut in G .

Proof. Clearly, any path between two terminals different from s_j would use at least one node from $M - \Gamma^{1/2}(S_j)$. Any path between terminals s_i and s_j uses a node of $\Gamma(S_i)$. Since $\Gamma(S_i)$ is disjoint from $\Gamma^{1/2}(S_j)$, such a path uses a node of $M - \Gamma^{1/2}(S_j)$. Hence $M - \Gamma^{1/2}(S_j)$ is a node multiway cut in G . $\square$

The weight of this node multiway cut can be bounded as follows.

$$\begin{aligned} W_M - W_{\Gamma^{1/2}(S_j)} &= W_{M^1} + W_{M^{1/2}} - W_{\Gamma^{1/2}(S_j)} \leq W_{M^1} + \left(1 - \frac{1}{k}\right) W_{M^{1/2}} \\ &= W_{M^1} + \frac{1}{2} \left(2 - \frac{2}{k}\right) W_{M^{1/2}} \leq \left(2 - \frac{2}{k}\right) \left(W_{M^1} + \frac{1}{2} W_{M^{1/2}}\right) \\ &= \left(2 - \frac{2}{k}\right) \text{maximum flow}. \quad \square \end{aligned}$$

The graph in Fig. 3 also shows that this approximate max-flow min-multiway cut theorem is tight. The maximum multicommodity flow for this instance is $k/2$ while the minimum node multiway cut has weight $k - 1$. This gives a ratio of $(2 - 2/k)$ between the minimum node multiway cut and the maximum flow.

3.4. Approximating the multiway cut

The ideas in Theorem 5 yield a polynomial time $(2 - 2/k)$ -approximation algorithm for the minimum node multiway cut problem. The algorithm is formalized in Fig. 4 and was the first algorithm to achieve an approximation guarantee better than the trivial k .

A slight modification to the example in Fig. 3 shows that the analysis of our algorithm is tight. The nodes with unit weight are now assigned a weight 2, while the center node has weight $k + \epsilon$. The center node is also the minimum node multiway cut. The optimal solution to the dual LP assigns a length label of $1/2$ to each node of weight 2 and a zero label to the center node. Hence, the multiway cut found by our algorithm has weight $2k - 2$ which is about $(2 - 2/k)$ times the weight of the optimum.

3.5. Non-half-integral example for directed case

We now give an example to show that for directed graphs, the LP-relaxation of the minimum multiway cut problem is not half-integral. We can again define a multicommodity flow problem by associating a commodity c_{ij} with each ordered pair of distinct terminals (s_i, s_j) ; terminal s_i is the source and s_j the sink for this commodity. The commodities are routed subject to conservation and capacity constraints with the objective of maximizing the sum of the flows of the commodities. This multicommodity flow problem when written as a linear program has a dual which can be viewed as an assignment of non-negative lengths to the edges, l_e , so that the length of any directed path between any two terminals is at least 1; the objective is to minimize $\sum_{e \in E} l_e w_e$. An integral solution to the dual LP is a multiway cut and hence the value of the maximum flow is a lower bound on the weight of the minimum multiway cut.

The optimal solution of this dual LP is not necessarily half-integral. Consider for example the graph of Fig. 5 with two terminals s_1, s_2 , where the arcs $\alpha_1 \rightarrow \alpha_2, \beta_2 \rightarrow \beta_1, \gamma_2 \rightarrow \gamma_1, \delta_1 \rightarrow \delta_2$ have weight 1 and the rest of the edges have large weight. The maximum flow is $4/3$, consisting of $2/3$ units of flow in each direction; the dual LP has a unique optimal (fractional) solution that assigns length $1/3$ to each of the four unit weight arcs and length 0 to the rest. The optimal (integral) 2-way cut in this case has weight 2.

This example for the edge-multiway cuts can be trivially transformed into an example for node multiway cuts by giving a large weight to each node and introducing a unit weight node to split each of the four unit weight edges.

Algorithm Node_multiway_cut;

1. Compute an optimal solution to the dual LP
 2. For each terminal s_i , compute S_i , $\Gamma(S_i)$ and $\Gamma^{1/2}(S_i)$
 3. Let j be such that $\Gamma^{1/2}(S_j)$ has maximum weight
 4. Output $\bigcup_{i=1}^k \Gamma(S_i) - \Gamma^{1/2}(S_j)$
- end.**

Fig. 4. Algorithm for approximating the minimum node multiway cut.

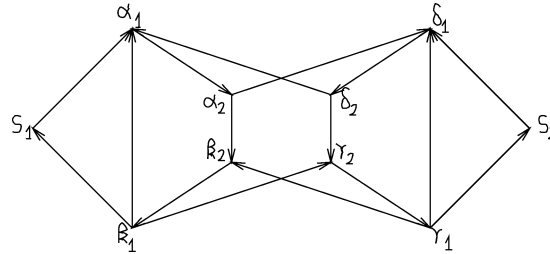


Fig. 5. The optimal dual solution is not half-integral.

4. Open problems

A number of interesting problems remain open in this area. In the case of planar undirected graphs with a fixed number of terminals, the edge multiway cut problem can be solved in polynomial time [4]. We do not know whether this result can be extended to the node version or to directed planar graphs.

Observe that the node cover problem reduces in an approximation-preserving manner to the undirected node multiway cut problem. Thus, improving the 2-approximation for the latter problem would also improve the approximation guarantee for the node cover problem (which is a long-standing open problem).

As mentioned in Section 2, the multiway cut problem is a special case of the multicut problem. The best approximation factors that are currently known for the general multicut problem are worse than the multiway case: $O(\log k)$ in the undirected (edge and node) case [7], and essentially the trivial $O(k)$ for directed graphs. On the negative side, we only know that the multicut problem is MAX SNP-hard—at least as hard as the node cover problem (even in the undirected edge case)—and so there is a gap in all cases, especially for directed graphs. Finally, a common generalization of the node multiway cut problem and the feedback node set problem, called the *subset feedback vertex set problem* was studied recently in [6], and a factor 8 approximation algorithm obtained. Given a special set of nodes in a node weighted undirected graph, this problem asks for a minimum weight set of nodes whose removal disconnects all cycles using any of the special nodes. It would be natural to expect the correct factor for this problem to be 2. In this context, we do not know if our half integrality theorem extends to an appropriate LP-relaxation of this problem. The edge version of this problem has a 2-approximation algorithm [5].

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